

19.

Given $U_4 = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}$

$$\frac{dU_4}{d\beta} = -\frac{1}{3} \frac{d}{d\beta} \left(\frac{\langle m^4 \rangle}{\langle m^2 \rangle^2} \right) = -\frac{1}{3} \left[\frac{d}{d\beta} (\langle m^4 \rangle) \frac{1}{\langle m^2 \rangle^2} - \frac{2\langle m^4 \rangle}{\langle m^2 \rangle^3} \frac{d}{d\beta} (\langle m^2 \rangle) \right]$$

$$\frac{d\langle m^4 \rangle}{d\beta} = \frac{d}{d\beta} \left(\frac{\sum m^4 e^{-\beta H}}{\sum e^{-\beta H}} \right) = \frac{\sum -m^4 H e^{-\beta H}}{\sum e^{-\beta H}} - \frac{\sum m^4 e^{-\beta H} \sum -H e^{-\beta H}}{(\sum e^{-\beta H})^2}$$

$= -\langle m^4 H \rangle + \langle m^4 \rangle \langle H \rangle$

$$\Rightarrow \frac{d\langle m^4 \rangle}{d\beta} = -\langle m^4 H \rangle + \langle m^4 \rangle \langle H \rangle$$

$$\frac{2\langle m^4 \rangle}{\langle m^2 \rangle^3} \frac{d\langle m^2 \rangle}{d\beta} = \frac{2\langle m^4 \rangle}{\langle m^2 \rangle^2} \frac{d \ln(\langle m^2 \rangle)}{d\beta}$$

$$\Rightarrow \frac{2\langle m^4 \rangle}{\langle m^2 \rangle^3} \frac{d\langle m^2 \rangle}{d\beta} = -\frac{2\langle m^4 \rangle \langle m^2 H \rangle}{\langle m^2 \rangle^3} + \frac{2\langle m^4 \rangle \langle H \rangle \langle m^2 \rangle}{\langle m^2 \rangle^3}$$

$$\Rightarrow \frac{dU_4}{d\beta} = -\frac{1}{3} \left[\frac{-\langle m^4 H \rangle + \langle m^4 \rangle \langle H \rangle}{\langle m^2 \rangle^2} - \frac{2\langle H \rangle \langle m^4 \rangle}{\langle m^2 \rangle^2} + \frac{2\langle m^4 \rangle \langle m^2 H \rangle}{\langle m^2 \rangle^3} \right]$$

$$\frac{dU_4}{d\beta} = -\frac{1}{3} \left[\frac{1}{\langle m^2 \rangle^2} \left[\frac{2\langle m^4 \rangle \langle m^2 H \rangle}{\langle m^2 \rangle} - \langle m^4 H \rangle - \langle H \rangle \langle m^4 \rangle \right] \right]$$

$$\Rightarrow \frac{dU_4}{d\beta} = -\frac{1}{3} \left[\frac{\langle m^4 \rangle}{\langle m^2 \rangle^2} \left[\frac{2\langle m^2 H \rangle}{\langle m^2 \rangle} - \frac{\langle m^4 H \rangle}{\langle m^4 \rangle} - \langle H \rangle \right] \right]$$

$$\frac{dU_4}{d\beta} = \frac{1}{3} \frac{\langle m^4 \rangle}{\langle m^2 \rangle^2} \left(\langle H \rangle - 2 \frac{\langle m^2 H \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 H \rangle}{\langle m^4 \rangle} \right)$$

$$\frac{1}{3} \frac{\langle m^2 \rangle}{\langle m^2 \rangle^2} = 1 - u_1$$

$$\Rightarrow \frac{d u_1}{d \beta} = (1 - u_1) \left(\langle H \rangle - 2 \frac{\langle m^2 H \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 H \rangle}{\langle m^2 \rangle} \right)$$

For a Hamiltonian $H = eV$ where e is the energy density and V the volume" ($V = L^2$)

$$\langle H \rangle = V \langle e \rangle$$

$$\langle H m^2 \rangle = V \langle A e \rangle$$

$$\Rightarrow \frac{d u_1}{d \beta} = V(1 - u_1) \left(\langle e \rangle - 2 \frac{\langle m^2 e \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 e \rangle}{\langle m^2 \rangle} \right)$$

For $u_2 = 1 - \frac{\langle m^2 \rangle}{3 \langle m^4 \rangle}$ the result follows the same form.

$$\frac{d u_2}{d \beta} = V(1 - u_2) \left(\langle e \rangle - \frac{2 \langle m^2 e \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 e \rangle}{\langle m^2 \rangle} \right)$$

b)

$$\frac{d \ln(\langle m^2 \rangle)}{d \beta} = \frac{d \langle m^2 \rangle}{d \beta} \frac{1}{\langle m^2 \rangle} = \frac{1}{\langle m^2 \rangle} \left(\langle m^2 \rangle \langle H \rangle - \langle m^2 H \rangle \right)$$

$$\frac{d \ln(\langle m^2 \rangle)}{d \beta} = \langle H \rangle - \frac{\langle m^2 H \rangle}{\langle m^2 \rangle} = V \left(\langle e \rangle - \frac{\langle m^2 e \rangle}{\langle m^2 \rangle} \right)$$

Similarly for $\langle m^4 \rangle$

$$\frac{d \ln \langle m^4 \rangle}{d \beta} = V \left(\langle e \rangle - \frac{\langle m^4 e \rangle}{\langle m^4 \rangle} \right)$$